Technische Universität Braunschweig



# QCN Presentation

Bloch Sphere

Entanglement, CHSH and MAC

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# Quantum State Tomography

Understanding Qubits and State Characterization



# Qubits and the Bloch Sphere (Pure States)

## What is a Qubit?

- Quantum counterpart of a classical bit.
- Described by a two-dimensional Hilbert space (H2).
- Represented as vectors, e.g., $|0\rangle$ and $|1\rangle$.

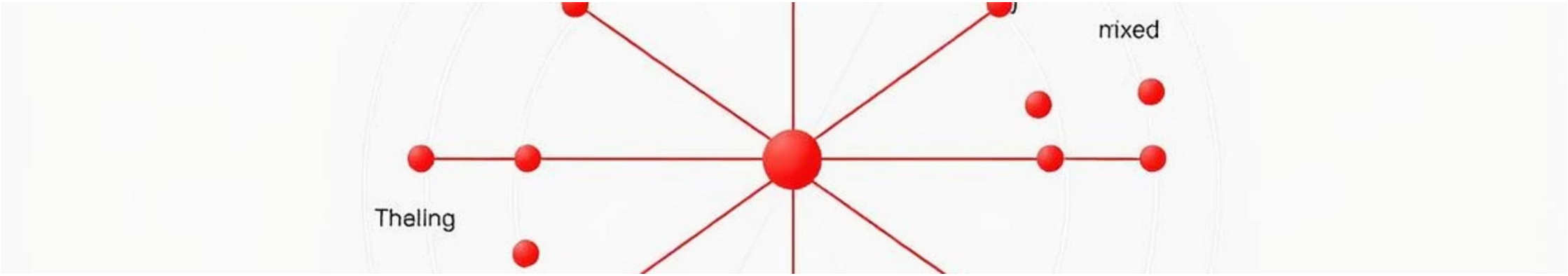
## Superposition

- Qubit exists in a combination of $|0\rangle$ and $|1\rangle$ simultaneously: $|\psi\rangle = c1|0\rangle + c2|1\rangle$.
- c1 and c2 are complex probability amplitudes.

## The Bloch Sphere (Pure States)

- Geometric representation of a qubit.
- Pure qubit states lie on the surface of a unit sphere.
- Parameterized by angles $\theta$ and $\phi$.






# Bloch Ball (Mixed States) and Purity

## Noisy Quantum States

- Difficult to perfectly prepare or measure quantum states.
- Lack of complete information leads to a "mixed quantum state."

## Density Operator Formalism

- Mathematical tool for pure and mixed states.
- Encompasses uncertainty from an ensemble of states.

## The Bloch Ball (Mixed States)

- Mixed qubit states correspond to points inside the Bloch sphere.
- Maximally mixed state is at the center (origin).

## Purity (P(ρ))

- Measures noisiness: $P(\rho) = Tr\{\rho^2\}$.
- Pure states: $P(\rho) = 1$.
- Mixed states: $P(\rho) < 1$.

# Pauli Matrices and Bloch Vector Representation

## Pauli Matrices

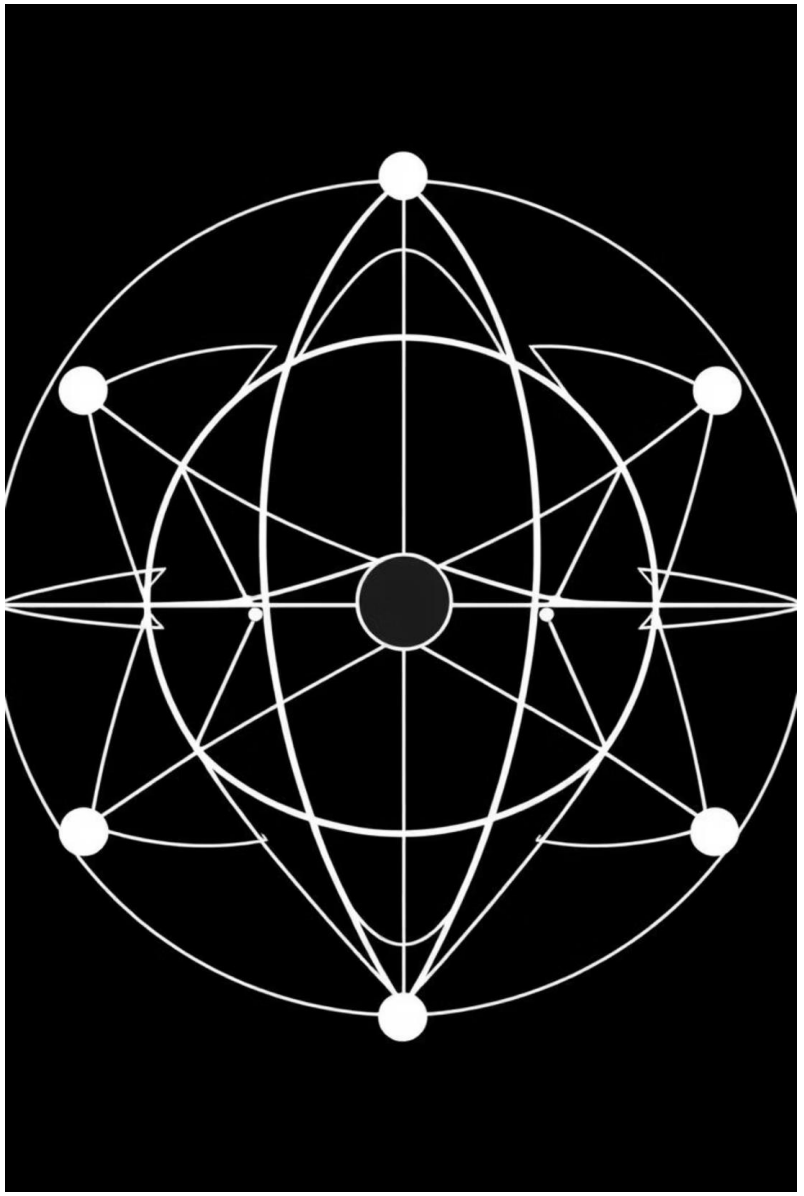
Fundamental operators (σx, σy, σz) for qubit Hilbert space.

- σx (X): Bit-flip
- σy (Y): Bit and phase-flip
- σz (Z): Phase-flip

## Bloch Vector (~r)

Represents any qubit density operator ρ: $\rho = (1/2) * (I + \tilde{r} \cdot \tilde{\sigma})$.

- Components: $r_i = Tr(\rho\sigma_i)$.
- Pure states: $|\tilde{r}| = 1$ (on surface).
- Mixed states: $|\tilde{r}| \leq 1$ (inside ball).



# Quantum State Tomography (QST)

## What is QST?

Technique to determine an unknown quantum state (density operator) from experimental measurement data.

## Why is QST Necessary?

- Single measurement doesn't reveal full state.
- Related to No-Cloning Theorem.
- Requires many identical copies for characterization.



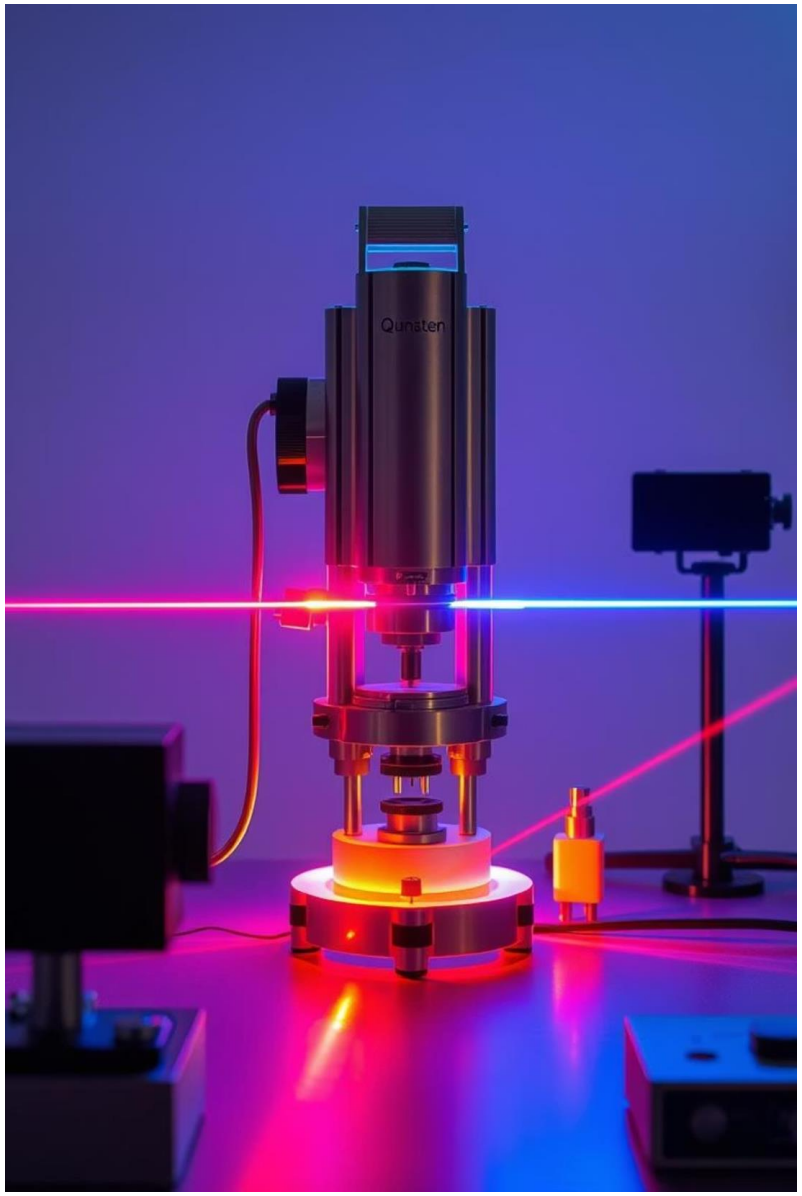
## Principle

Density operator determined from expected values of a complete complete set of observables (e.g., Pauli matrices).



## Process

Repeated measurements in different bases statistically estimate expectation values, which define the Bloch vector.



# QST: Practical Steps for a 1-Qubit System

### 1. Prepare Ensemble

Create many identical copies of the unknown 1-qubit state for statistical precision.

### 2. Measure in Different Bases

Divide copies into groups and perform projective measurements in Z-basis, X-basis, and Y-basis. and Y-basis.

### 3. Calculate Expected Values

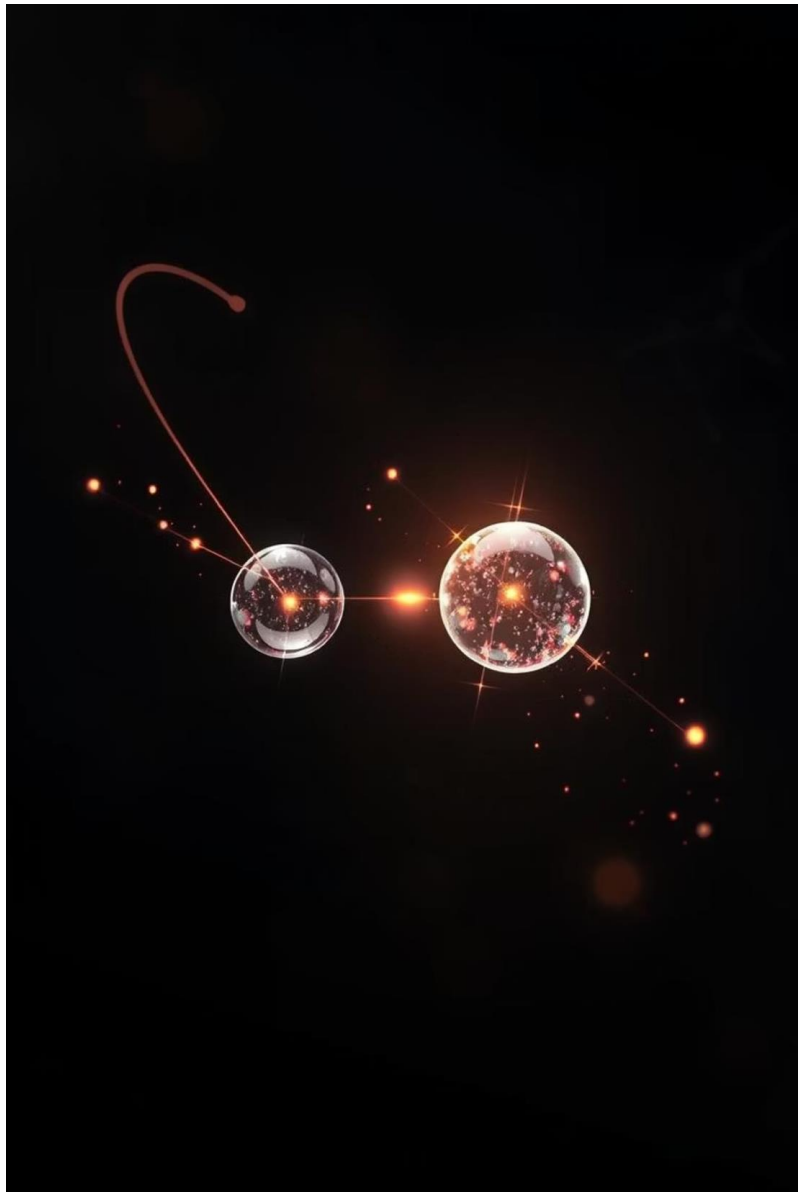
From measurement outcomes, compute $\langle\sigma_x\rangle_\rho$, $\langle\sigma_y\rangle_\rho$, and $\langle\sigma z\rangle_\rho$, which are the Bloch vector vector components.

### 4. Reconstruct Density Operator

Use the determined Bloch vector (~r) to reconstruct $\rho = (1/2) * (I + r_x\sigma_x + r_y\sigma_y + rz\sigma z)$. $rz\sigma z)$.

# Entanglement: The Unique Quantum Resource

Explore entanglement's role in quantum computing and communication, highlighting its non-local correlations and power.



# The CHSH Game: Setup & Rules



## Purpose

Demonstrates entanglement's power, separating classical and quantum physics.



## Participants

Alice (A), Bob (B), and a Referee (R), spatially separated.



## Inputs

Referee sends random binary inputs (x, y) to Alice and Bob.

Bob.



## Winning Condition

Alice and Bob win if $(x \wedge y) = (a \oplus b)$.

# CHSH Game: Classical Strategy and Limitations

## Optimal Classical Strategy

Alice and Bob always output a=0 and b=0.

## Winning Probability

They win 3 out of 4 times, achieving a maximal winning probability of probability of 75%.

| x | y | $x \wedge y$ | $a \oplus b$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 |

Shared classical randomness does not allow exceeding this 75% limit.

# CHSH Game: Quantum Strategy and Advantage



### Quantum Resource

Alice and Bob share an entangled quantum state (e.g., Bell state).



### Local Measurements

They perform local measurements based on their input bits.



### Increased Win Rate

Achieve ~85.4% winning probability ($\cos^2(\pi/8)$).



### Quantum Advantage

~10% increase highlights entanglement's superiority over classical classical correlations.

# Entanglement as a Powerful Resource in Communication



### Super-dense Coding

Transmit two classical bits by sending one qubit, doubling doubling channel capacity and ensuring privacy.



### Quantum Teleportation

Transmit an unknown quantum state using pre-shared shared entanglement and two classical bits.

These protocols fundamentally improve communication performance, enhancing data rates and security.

# Multiple-Access Channel (MAC): A Communication Model

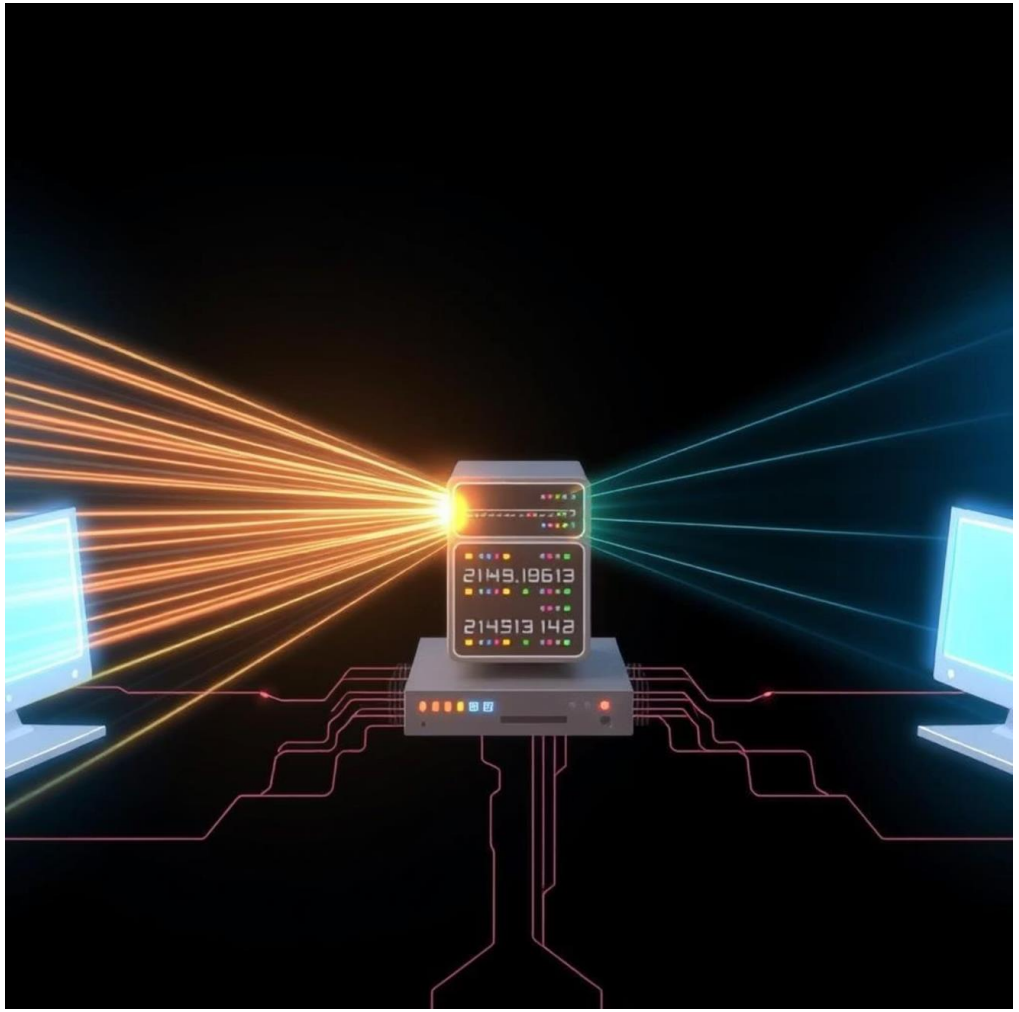
## Definition

Multiple senders transmit simultaneously to a single receiver via a common channel.

## Operational Principles

- Message encoding into codewords.
- Synchronized parallel transmission.
- Channel interference from combined inputs.
- Receiver decodes noisy, combined output.

Goal: Minimize decoding error and maximize achievable communication rates.



# CHSH Game & MAC: Demonstrating Quantum Enhancement




## Classical MAC Performance

Limited by 75% success probability, introducing noise.

## Quantum-Enhanced MAC

Increased success probability (approx. 85%) leads to enhanced MAC performance.

This demonstrates how quantum communications exceed classical limits by exploiting entanglement, leading to lower Bit Error Rates (BER) and higher capacities.

# Conclusion: The Future of Quantum Communication Networks

### Entanglement's Role

Fundamental for capabilities beyond classical resources.

### CHSH Benchmark

Quantifies quantum strategy superiority, highlighting entanglement's power.

### Practical Application

CHSH principles enhance MAC performance, improving communication systems.

### Future Outlook

Crucial for developing quantum communication networks, networks, despite challenges.

# Thank you!

Questions?